

MPS Settings Strategy: Preliminary Heuristics

lappy

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Abstract

This document develops a principled basis for choosing **lappy** solver settings—primarily **n_basis**, **n_fb**, **n_fs**, **rtol**, **ltol**, and **ppl**—as functions of the desired accuracy, the number of eigenvalues requested, and the geometry of the domain. The analysis is preliminary; the benchmark sweeps are designed to refine the constants and validate the functional forms.

Contents

1 Background: the MPS tension function

The Method of Particular Solutions (MPS) finds Laplacian eigenvalues as approximate zeros of a *tension function* $\sigma(\lambda)$. Given a basis of N particular solutions $\{\varphi_1, \dots, \varphi_N\}$ —exact solutions to $(\Delta + \lambda)u = 0$ in $\mathbb{R}^2$ —the tension at λ is

$$\sigma(\lambda) = \min_{\|c\|=1} \frac{\|\sum_i c_i \varphi_i\|_{L^2(\partial\Omega)}}{\|\sum_i c_i \varphi_i\|_{L^2(\Omega)}}. \quad (1)$$

$\sigma(\lambda)$ is small when there exists a linear combination that nearly vanishes on $\partial\Omega$ while remaining non-trivial in the interior, i.e. when λ is near a true Dirichlet eigenvalue. In the GSVD formulation used here, $\sigma(\lambda)$ equals the smallest generalised singular value of the pair (boundary evaluation matrix, interior evaluation matrix).

The eigenvalue error is bounded by the tension:

$$\frac{|\hat{\lambda} - \lambda|}{\lambda} \lesssim C \sigma(\hat{\lambda}), \quad (2)$$

where C depends on the local spectral gap. Achieving relative accuracy **ltol** therefore requires $\sigma(\lambda) \lesssim \mathbf{ltol}$.

2 Why convergence is exponential in N

The key theoretical fact underlying MPS is that the tension decays *exponentially* in N for a well-chosen basis,

$$\sigma(\lambda) \sim \exp(-\gamma N), \quad (3)$$

where $\gamma > 0$ is a geometry-dependent convergence rate. This follows from approximation theory for analytic functions.

Fourier–Bessel (FB) basis. At a corner with interior angle α , the Dirichlet eigenfunctions have the form

$$u(r, \theta) = \sum_{n \geq 1} a_n r^{n\pi/\alpha} \sin\left(\frac{n\pi\theta}{\alpha}\right) \quad (4)$$

in local polar coordinates (r, θ) centred at the corner. This is the *exact* singular form dictated by the geometry. The FB basis at that corner consists precisely of $J_{n\pi/\alpha}(\sqrt{\lambda}r) \sin(n\pi\theta/\alpha)$, which matches this structure term by term. The residual after including N terms is therefore the tail of an analytic series,

$$\|u - u_N\| \sim C \exp(-\gamma_{\text{corner}} N), \quad (5)$$

where γ_{corner} is determined by the analyticity radius of u away from that corner. Since the eigenfunction is analytic everywhere except at corners, and the FB basis handles the corner singularities exactly, the remaining boundary residual falls off exponentially fast.

Fundamental Solution (FS) basis. Sources placed at distance d outside the boundary generate functions that are analytic on all of $\bar{\Omega}$. For a smooth domain (or as a supplement to FB functions on a polygon), N such sources achieve

$$\|u - u_N\| \sim C \exp(-\gamma_{\text{fs}} N), \quad (6)$$

where $\gamma_{\text{fs}} \sim \pi d/L$ for sources distributed at distance d from a boundary of length L . The convergence rate decreases as $d \rightarrow 0$, so source placement matters.

Consequence for basis sizing. The tension target `ltol` requires

$$N \gtrsim \frac{|\log(\text{ltol})|}{\gamma}. \quad (7)$$

This is a lower bound from the accuracy requirement alone. As we will see, the oscillation count often dominates in practice.

3 The oscillation count argument

The n -th eigenfunction oscillates at spatial frequency $\sqrt{\lambda_n}$. Along a boundary segment of length ℓ it executes roughly $\ell\sqrt{\lambda_n}/(2\pi)$ oscillations. A basis of N functions centred at a single point can accurately represent at most $O(N)$ oscillation modes. To capture the first

k eigenvalues across all boundary segments we need the basis to span the oscillations at scale $\sqrt{\lambda_k}$. Summing over all corners or source locations,

$$N_{\text{osc}} \sim C \frac{P\sqrt{\lambda_k}}{2\pi}, \quad (8)$$

where P is the perimeter. Substituting Weyl's law $\lambda_k \sim 4\pi k/A$ gives $\sqrt{\lambda_k} \sim 2\sqrt{\pi k/A}$, and hence

$$N_{\text{osc}} \sim C \frac{P}{\sqrt{A}} \sqrt{k}. \quad (9)$$

The factor $P/\sqrt{A}$ is the *isoperimetric ratio*. It equals $2\sqrt{\pi} \approx 3.54$ for a circle and grows for elongated or geometrically complex domains. Representative values: a square has $P/\sqrt{A} = 4$; an L -shape has $P/\sqrt{A} \approx 5$ –6; the GWW isospectral domains have $P/\sqrt{A} \approx 7$ –9.

4 The combined N_{basis} heuristic

The two constraints—accuracy (??) and oscillation count (??)—combine as

$$N_{\text{basis}} \sim C \frac{P}{\sqrt{A}} \sqrt{k} F_{\text{geom}} f(\text{ltol}), \quad (10)$$

where we now describe each factor.

4.1 Accuracy factor $f(\text{ltol})$

Once N exceeds the oscillation count, the tension drops rapidly. Because convergence is exponential, each additional basis function buys a fixed multiplicative improvement. A reasonable approximation is

$$f(\text{ltol}) = \max\left(1, \alpha |\log(\text{ltol})|^\beta\right), \quad \beta \in [0.5, 1.0].$$

The benchmark data shows that going from $\text{ltol} = 10^{-10}$ to 10^{-16} requires only 30–50% more basis functions, consistent with $\beta \approx 0.5$. We conservatively use $\beta = 1$ (linear in $|\log(\text{ltol})|$) until more data are available.

4.2 Geometric difficulty factor F_{geom}

Re-entrant corners ($\alpha > \pi$). The singularity exponent $\pi/\alpha < 1$ means the eigenfunction is less smooth at the corner; more FB terms are needed to resolve it. A rough per-corner factor is $(\alpha/\pi)^{1/2}$ for $\alpha > \pi$, giving

$$F_{\text{corners}} = \prod_{\substack{i: \text{corner} \\ \alpha_i > \pi}} \left(\frac{\alpha_i}{\pi}\right)^{1/2}. \quad (11)$$

For example, the L -shape has one 270° corner ($\alpha = 3\pi/2$), giving $F_{\text{corners}} = \sqrt{3/2} \approx 1.22$. This is consistent with needing $N \approx 120$ for the L -shape versus $N \approx 60$ –80 for rectangles and triangles.

Irrational corner angles. If π/α_i is irrational, the required FB orders are non-integer-order Bessel functions with slower convergence than integer-order ones. The disk sector at $\theta = \pi\sqrt{2}/2$ is the clearest example in the benchmark data; it struggles even at $N_{\text{basis}} = 160$. For irrational angles the convergence rate γ is degraded and a larger empirical multiplier is needed.

Smooth curved boundaries. Domains with curved sides but no corners (disk, ellipse, stadium) require no FB functions. The FS convergence rate depends on the source placement distance d relative to the inradius r_{in} : $\gamma_{\text{fs}} \sim \pi d/r_{\text{in}}/P_{\text{norm}}$. A source distance of $d \approx 0.15 r_{\text{in}}$ is the current default; the `fs_placement` sweep will quantify the optimal value.

4.3 Summary and comparison with data

Putting it together, the working preliminary heuristic is

$$N_{\text{basis}} \approx \text{round}\left(C \frac{P}{\sqrt{A}} \sqrt{k} F_{\text{geom}} |\log_{10}(\text{ltol})|\right), \quad (12)$$

with $C \approx 2\text{--}4$ to be fit from the benchmark data. Table ?? compares predictions to preliminary observations.

Domain	$P/\sqrt{A}$	F_{geom}	Predicted N ($k = 10$, $\text{ltol} = 10^{-14}$)	Observed N
rect (2×1)	4.2	1.0	55–70	≈ 60
iso_right_tri	4.6	1.0	60–75	≈ 70
eq_tri	4.6	1.0	60–75	≈ 60
L_shape	5.5	1.22	85–110	≈ 120
GW1	≈ 8	≈ 1.5	150–200	> 160 (tbc)

Table 1: Predicted versus observed basis sizes required for $k = 10$ eigenvalues at relative accuracy $\text{ltol} = 10^{-14}$. Predictions use (??) with $C = 3$.

5 Role of k (number of requested eigenvalues)

The number of requested eigenvalues k enters through the $\sqrt{k}$ scaling of the oscillation count in (??). Key implications:

Sublinear cost growth. Doubling k requires only $\sqrt{2} \approx 41\%$ more basis functions, not double. This is one of the attractive properties of MPS relative to finite element methods, which typically scale as $O(k)$ or worse in practice.

Sizing for the largest eigenvalue. The basis must be sized for λ_k , the *largest* eigenvalue requested. Using a basis sized for $k = 10$ when $k = 25$ eigenvalues are requested will reliably find the first few but miss or inaccurately compute the upper range. This is open issue 2.7 (adaptive basis sizing): currently `evp.solve(k)` uses whatever basis was constructed at initialization regardless of k .

Practical lower bound.

$$N_{\text{basis}} \geq N_{\text{min}}(k) = \text{round}\left(C \frac{P}{\sqrt{A}} \sqrt{k} F_{\text{geom}}\right), \quad (13)$$

with $C \approx 4\text{--}6$ as a conservative starting point. The GSVD cost scales as $O(N^2 n_{\text{pts}})$, so there is a penalty for over-sizing, but it is not catastrophic for moderate N .

Interaction with `ppl`. The `ppl` parameter (points per level in the bracket search) also interacts with k . The solver searches for k tension minima on $[\lambda_{\text{min}}, \lambda_{\text{max}}]$. With too few points per level (`ppl`= 5), eigenvalues can be missed entirely—this is the primary cause of solver failures in the benchmark data at small N_{basis} . The recommended minimum is `ppl`= 10; for difficult domains or large k , `ppl`= 15–20 is safer.

6 The FB/FS split

Given a total budget of N_{basis} functions, how should they be divided between Fourier-Bessel (`n_fb`) and Fundamental Solutions (`n_fs`)?

Pure polygon, no curved sides. FB only (`n_fs`= 0). The FB basis exactly matches the singularity structure; FS functions add smooth global corrections that are redundant once the FB orders are high enough.

Pure smooth domain (no corners). FS only (`n_fb`= 0). There is no singularity structure for the FB basis to capture.

Mixed domain (polygon with curved sides). Use both. FB functions handle corner singularities; FS functions supplement the smooth regions.

FB order allocation across corners. With n_{fb} functions distributed across n_c corners, the `fb_strategy` parameter controls allocation:

- **singular_angle_weighted** (default): allocates more orders to corners with larger (more re-entrant) angles. This is theoretically motivated since more singular corners require more FB terms.
- **uniform**: equal orders per corner. Suboptimal when corner angles vary widely.
- **singular_only**: all orders at non-regular corners (those where π/α is not an integer), zero at regular corners. Aggressive but correct in theory for polygons with a mix of regular and singular corners.

The `fb_corner_sweep` will quantify the accuracy difference between strategies. The preliminary expectation is that for domains with one dominant singular corner (such as the L -shape), `singular_only` and `singular_angle_weighted` will outperform `uniform`.

7 Collocation points

The boundary and interior collocation points affect accuracy and stability, but are less sensitive than basis size.

Boundary points (`bdry_pts_factor`). The linear system is overdetermined: we use $\text{bdry_pts_factor} \times N_{\text{basis}}$ boundary points. A factor of 2.0 (the default) is standard for MPS and gives good conditioning. Going below 1.5 risks an underdetermined system; going above 3.0 adds cost without meaningful benefit. Points should be distributed proportionally to segment length. Gauss–Legendre quadrature points are preferable to uniform spacing because they avoid endpoint clustering artifacts.

Interior points (`int_pts_factor`). Used for basis normalisation (L^2 norm over Ω). A factor of 1.0 is typically sufficient; the normalisation is not sensitive to this choice as long as the points are reasonably distributed.

8 Regularisation: `rtol` and `ltol`

The GSVD-based MPS involves two tolerance parameters with different roles.

`rtol` (regularisation tolerance). Singular values of the basis matrix below $\text{rtol} \times \sigma_{\max}$ are discarded. This controls ill-conditioning from near-linearly-dependent basis functions.

- Too small: numerical noise dominates and the tension function becomes noisy, causing missed or spurious eigenvalues.
- Too large: legitimate but small singular values (corresponding to weakly excited modes) are discarded, causing missed eigenvalues.

Preliminary results from the `regularization_sweep` suggest the sweet spot is $\text{rtol} \in [10^{-12}, 10^{-14}]$ for well-conditioned domains. The GWW domain degrades significantly below $\text{rtol} = 10^{-12}$, suggesting the basis becomes ill-conditioned at large N for complex geometries.

A geometry-motivated heuristic is

$$\text{rtol} \sim C_r \exp(-\gamma N_{\text{basis}}), \quad (14)$$

i.e., set `rtol` to the expected tension floor for the given basis size and geometry. In practice $\text{rtol} = 10^{-14}$ is a safe default; tighten to 10^{-16} only if the basis is verified to be well-conditioned.

`ltol` (eigenvalue acceptance threshold). A tension minimum is accepted as an eigenvalue only if $\sigma(\hat{\lambda}) < \text{ltol}$. This should be set relative to the accuracy the basis is expected to achieve:

$$\text{ltol} \sim 10 \times \exp(-\gamma N_{\text{basis}}) \sim 10 \times \sigma_{\text{floor}}(N_{\text{basis}}). \quad (15)$$

Setting `ltol` too tight causes real eigenvalues to be rejected; too loose accepts noise peaks as eigenvalues. The current default of `ltol = 10-15` is aggressive; for small bases ($N < 50$) or complex domains, `ltol = 10-10–10-12` is more reliable.

The `rtol`/`ltol` relationship. Both tolerances should be set consistently. A rough guideline is

$$\text{ltol} \approx 100 \times \text{rtol}, \quad (16)$$

which ensures that basis regularisation does not create artificial tension minima above the eigenvalue acceptance threshold.

9 Practical recommendations (preliminary)

Until the benchmark sweeps are complete, the following settings are conservative starting points.

Step 1: estimate N_{basis} .

```
import numpy as np

def estimate_n_basis(domain, k, ltol=1e-14):
    P = domain.perim
    A = domain.area
    iso_ratio = P / np.sqrt(A)
    # geometric difficulty from re-entrant corners
    angles = domain.int_angles
    F_geom = np.prod([np.sqrt(a / np.pi) for a in angles if a > np.pi]) or 1.0
    C = 5.0 # conservative constant; refine from benchmarks
    return round(C * iso_ratio * np.sqrt(k) * F_geom * abs(np.log10(ltol)))
```

Step 2: set the FB/FS split.

- Polygon with all straight sides: `n_fs=0`, `n_fb=  $N_{\text{basis}}$` .
- Smooth domain: `n_fb=0`, `n_fs=  $N_{\text{basis}}$` .
- Mixed: `n_fb=round(0.5  $N_{\text{basis}}$ )`, `n_fs=  $N_{\text{basis}}$  - n_fb`.

Use case	rtol	ltol	ppl
Quick / exploratory	10^{-10}	10^{-8}	10
Standard	10^{-12}	10^{-10}	10
High accuracy	10^{-14}	10^{-12}	15
Near-degenerate eigenvalues	10^{-14}	10^{-12}	20

Table 2: Recommended tolerance settings by use case.

Step 3: set tolerances.

Step 4: check the result. A result should be treated as reliable if:

- `len(eigs) == k` (all requested eigenvalues found),
- `max(tensions) < 1tol` (all eigenvalues accepted with margin),
- `median(rel_errors) < 1tol` (when reference eigenvalues are available).

If `len(eigs) < k`, increase N_{basis} by 25–50% and/or increase `ppl`.

10 Open questions for the benchmark data

The following will be addressed as more sweep results become available.

1. **The constant C .** The current range $C \approx 2\text{--}6$ is too wide. The `basis_size` sweep will pin this down per domain class.
2. **The accuracy exponent β .** Is $f(1tol) \sim |\log(1tol)|$ or $\sim |\log(1tol)|^{1/2}$? The `regularization_sweep` data will answer this.
3. **Optimal FB/FS ratio for mixed domains.** The `basis_size` sweep covers `fb_fraction` $\in \{0, 0.25, 0.5, 0.75, 1.0\}$; we expect a clear optimum around 0.5–0.75 for polygons.
4. **FS source distance scaling.** The `fs_placement` sweep will determine how accuracy depends on `fs_d_scale`. The current theoretical prediction is $\gamma_{\text{fs}} \sim \pi d/r_{\text{in}}$; the data will test this.
5. **The irrational angle case.** The disk sector at $\theta = \pi\sqrt{2}/2$ is a red flag. We need to understand whether this is a basis sizing issue, an FB order allocation issue, or a fundamental limitation of the FB convergence rate.
6. **The $\sqrt{k}$ scaling exponent.** The $\sqrt{k}$ prediction from Weyl’s law should be directly testable from the `n_eigs` sweep.